POLY PMNA · DIPLOMA CURRICULUM REVISION 2026

Course 2009B | Semester I & II | Lab | 1.5 Credits

# Engineering Workshop Practice

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

| | |
|---|---|
| Programmes | **Computing, electronics and allied programme group listed in the official syllabus** |
| Contact hours | **45 + 45** |
| Teaching scheme | **0:0:3:0** |
| Credits | **1.5** |
| CIE / ESE | **Practical CIE 60 / Practical ESE 40** |
| Self-learning | **1.5 h/week; 45 h total** |

CURRICULUM INTEGRITY

## Official Source Declaration and Verified Inconsistencies

This handbook is based only on the supplied official **Diploma Curriculum Revision 2026** syllabus PDF for Course **2009B**, titled **Engineering Workshop Practice**. Objectives, course outcomes, teaching scheme, major topics, detailed coverage, activities and assessment are drawn from that source.

**Source treatment:** PDF table wrapping can interrupt a heading across lines. In such cases, this handbook retains the official intent in a clearly labelled topic. Normal teaching explanations, study flows, diagrams and Malayalam-support boxes are handbook support rather than claims of official wording.

**Verified source note:** No web research was used to establish syllabus content. Official references and URLs appear only where they are listed in the supplied PDF.

COURSE DASHBOARD

## What You Are Expected to Learn

**45 + 45**

Contact hours

**1.5**

Credits

**Practical CIE 60**

CIE

**Practical ESE 40**

ESE

### Course introduction

Engineering Workshop Practice establishes safe, accurate and collaborative habits in carpentry, fitting, welding, sheet-metal work, electrical wiring and basic electronics. Evaluation emphasises practical execution, documentation, record quality and an open-ended experiment.

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### Prerequisite foundation

Trigonometry/basic mathematics, electrical-conductivity/electronics basics, and Engineering Drawing with CAD (1004).

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

**Teaching and assessment scheme**

| L | T | P | S | Self-learning/week | Contact hours | Credits | CIE | ESE |
|---|---|---|---|---|---|---|---|---|
| 0 | 0 | 3 | 0 | 1.5 h/week; 45 h total | 45 + 45 | 1.5 | Practical CIE 60 | Practical ESE 40 |

STUDY METHOD

## How to Use This Handbook

### 1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

### 2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

### 3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

### 4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

# Objectives and Course Outcomes

## Official course objectives

1. Familiarise workplace safety and fabrication drawings.
2. Identify and use measuring, marking, holding, striking and cutting tools.
3. Practise electrical wiring and soldering.
4. Operate machines, power tools and workshop equipment safely.

## Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

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**Official Course Outcomes**

| CO | Official outcome | Bloom level |
|---|---|---|
| CO1 | Apply safety, 5S and carpentry tools/materials to make wooden boxes and forms for electronic applications. | Apply |
| CO2 | Use fitting-shop tools and precision instruments to make clamps, brackets, heat sinks and related parts. | Apply |
| CO3 | Use welding-shop tools to make joints for casings, racks and switchgear-related fabrications. | Apply |
| CO4 | Use sheet-metal tools and measuring instruments to fabricate electronic-equipment enclosures. | Apply |
| CO5 | Practise electrical wiring, motor connection and soldering of basic electronic circuits. | Apply |

| Official CO-PO articulation matrix | | | | | | | |
|---|---|---|---|---|---|---|---|
| **CO** | **PO1** | **PO2** | **PO3** | **PO4** | **PO5** | **PO6** | **PO7–PO11** |
| CO1 | 3 | 2 | 2 | 2 | 2 | – | – |
| CO2 | 3 | 2 | 2 | 2 | 2 | – | – |
| CO3 | 3 | 2 | 2 | 2 | 2 | – | – |
| CO4 | 3 | 2 | 2 | 2 | 2 | – | – |
| CO5 | 3 | 2 | 2 | 2 | 2 | – | – |

SYLLABUS COVERAGE

# Curriculum Coverage Map

| All official major topics mapped to handbook chapters | | | | | |
|---|---|---|---|---|---|
| **Module** | **Title** | **Official major topics** | **Hours** | **CO** | **Handbook section** |
| 1 | Industrial Safety and Fire Practice | Industrial safety, hazardous situations, personal protection, fire classes and hands-on fire-extinguisher practice. | Safety induction | CO1 | Module 1 |
| 2 | Carpentry and Wood Alternatives | Woodworking tools; MDF, WPC and plywood; marking, planing, cutting, chiselling and grooving; wooden-panel fabrication. | 18 practical hours | CO1 | Module 2 |
| 3 | Fitting Shop | Fitting tools and operations; heat sinks, clamps and brackets; drilling, screwing and precision measurement. | 18 practical hours | CO2 | Module 3 |
| 4 | Welding Shop | Welding safety; equipment and PPE; straight-line/spot welding; joining simple cabinets. | 18 practical hours | CO3 | Module 4 |
| 5 | Sheet Metal Work | Sheet-metal tools; cutting, bending, edging; fasteners; fabrication of casings for power supplies, inverters and UPS. | 18 practical hours | CO4 | Module 5 |
| 6 | Electrical Wiring and Basic Electronics | Electrical safety; one-way/staircase wiring; DOL starter; passive/active parts; rectifier/filter and Arduino familiarisation; soldering. | 18 practical hours | CO5 | Module 6 |

LEARNING ROADMAP

# How the Modules Connect

### Industrial Safety and Fire Practice

Industrial safety, hazardous situations, personal protection, fire classes and hands-on fire-extinguisher practice.

CO1 · Safety induction

### Carpentry and Wood Alternatives

Woodworking tools; MDF, WPC and plywood; marking, planing, cutting, chiselling and grooving; wooden-panel fabrication.

CO1 · 18 practical hours

### Fitting Shop

Fitting tools and operations; heat sinks, clamps and brackets; drilling, screwing and precision measurement.

CO2 · 18 practical hours

### Welding Shop

Welding safety; equipment and PPE; straight-line/spot welding; joining simple cabinets.

CO3 · 18 practical hours

### Sheet Metal Work

Sheet-metal tools; cutting, bending, edging; fasteners; fabrication of casings for power supplies, inverters and UPS.

CO4 · 18 practical hours

### Electrical Wiring and Basic Electronics

Electrical safety; one-way/staircase wiring; DOL starter; passive/active parts; rectifier/filter and Arduino familiarisation; soldering.

CO5 · 18 practical hours

MODULE 1

# Industrial Safety and Fire Practice

Industrial safety, hazardous situations, personal protection, fire classes and hands-on fire-extinguisher practice.

Safety induction | CO1 | Understand · Apply

### Learning goals

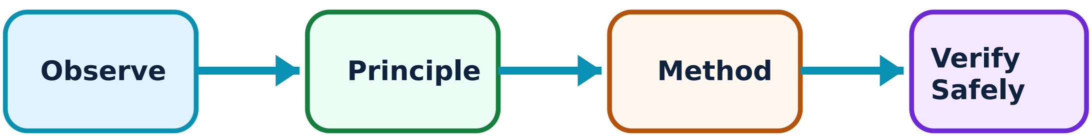
Identify core terms, explain the principle and complete the stated application or practical outcome.

### Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

### Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.




Study flow for Industrial Safety and Fire Practice: identify the system, state its principle, follow the correct method and verify the result safely.

### 1. Safe workshop behaviour and 5S−

Safe work begins before machine start: assess job, wear appropriate PPE, inspect tools, keep walkways clear and follow authorised procedures. 5S supports sorting, arranging, cleaning, standardising and sustaining disciplined workspace.

### Study and application method

Pause before work to identify hazard, control and emergency response. Return tools to correct location and report damaged equipment.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

### Practical method

1. Read the aim and identify apparatus, materials and workspace controls.
2. Prepare a labelled diagram, tabulation sheet or job plan before execution.
3. Carry out the prescribed operation methodically and record observations while working.
4. Verify the result against the aim, clean the work area and complete the record.

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**Common mistake / precaution:** Never operate equipment after removing guard or while distracted.

## 2. Fire classes and extinguishers−

Fire control depends on fuel type and extinguishing method. Official syllabus requires recognition of fire classes, matching extinguisher use and supervised hands-on practice.

### Study and application method

Raise alarm, follow local instruction, maintain safe escape route and use extinguisher only when trained and fire is small/controllable.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

### Practical method

1. Read the aim and identify apparatus, materials and workspace controls.
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**Common mistake / precaution:** Do not use water indiscriminately on electrical or flammable-liquid fires.

**Module 1 revision cue:** Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

## Carpentry and Wood Alternatives

Woodworking tools; MDF, WPC and plywood; marking, planing, cutting, chiselling and grooving; wooden-panel fabrication.

**18 practical hours** **CO1** **Understand · Apply**

### Learning goals

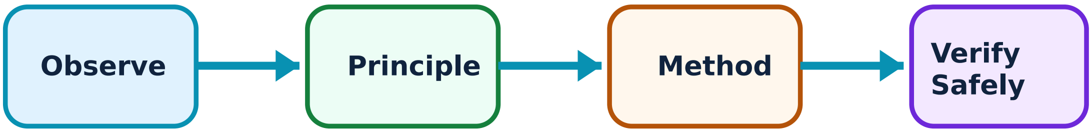
Identify core terms, explain the principle and complete the stated application or practical outcome.

### Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

### Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.




Study flow for Carpentry and Wood Alternatives: identify the system, state its principle, follow the correct method and verify the result safely.

## 1. Tools, materials and preparation−

Carpentry accuracy depends on selecting measuring, marking, cutting, holding and striking tools for intended operation. MDF, WPC and plywood are official alternative materials and differ from solid timber in working behaviour.

### Study and application method

Study board orientation, mark from datum edge and secure work before cutting. Record tool name, use and safety precaution.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

### Practical method

1. Read the aim and identify apparatus, materials and workspace controls.
2. Prepare a labelled diagram, tabulation sheet or job plan before execution.
3. Carry out the prescribed operation methodically and record observations while working.
4. Verify the result against the aim, clean the work area and complete the record.

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**Common mistake / precaution:** Do not cut unsupported work or place hands in saw line.

## 2. Wood-panel box fabrication−

The official activity develops skill in cutting/drilling panels, making a simple box, and fixing hinges/handles. Accuracy comes from clear drawing, square assembly and final inspection.

### Study and application method

Sequence drawing, measurement, marking, cutting, trial fit, drilling, fastening, inspection and finish. Check diagonals/corner squareness before final fastening.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

### Practical method

1. Read the aim and identify apparatus, materials and workspace controls.
2. Prepare a labelled diagram, tabulation sheet or job plan before execution.
3. Carry out the prescribed operation methodically and record observations while working.
4. Verify the result against the aim, clean the work area and complete the record.

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**Common mistake / precaution:** Power tools require eye protection, correct clamping and instructor-approved settings.

**Module 2 revision cue:** Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

# Fitting Shop

Fitting tools and operations; heat sinks, clamps and brackets; drilling, screwing and precision measurement.

18 practical hours | CO2 | Understand · Apply

### Learning goals

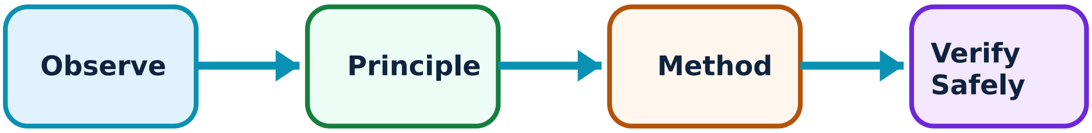
Identify core terms, explain the principle and complete the stated application or practical outcome.

### Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

### Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.




Study flow for Fitting Shop: identify the system, state its principle, follow the correct method and verify the result safely.

## 1. Fitting tools and measurement−

Fitting uses measured metal removal by marking, cutting, filing, drilling and finishing. Vernier calipers and micrometers are precision instruments, not rough layout tools.

### Study and application method

Read zero, hold instrument square to dimension and avoid overtightening. Mark from datum and centre-punch drilling positions.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

### Practical method

1. Read the aim and identify apparatus, materials and workspace controls.
2. Prepare a labelled diagram, tabulation sheet or job plan before execution.
3. Carry out the prescribed operation methodically and record observations while working.
4. Verify the result against the aim, clean the work area and complete the record.

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**Common mistake / precaution:** A file must have fitted handle; never use a file as lever or strike it with hammer.

### 2. Heat-sink, clamp and bracket fabrication−

Official applications require aluminium heat sinks or fabrication clamps/brackets, then drilling, screwing and fixing. Parts need functional dimensions and acceptable finish.

#### Study and application method

Use drawing for hole positions, edge distances and bends. Deburr every sharp edge and verify fit against intended panel before submission.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

#### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

#### Practical method

1. Read the aim and identify apparatus, materials and workspace controls.
2. Prepare a labelled diagram, tabulation sheet or job plan before execution.
3. Carry out the prescribed operation methodically and record observations while working.
4. Verify the result against the aim, clean the work area and complete the record.

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**Common mistake / precaution:** Freshly cut or ground metal edges can cause injury.

**Module 3 revision cue:** Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

# Welding Shop

Welding safety; equipment and PPE; straight-line/spot welding; joining simple cabinets.

18 practical hours | CO3 | Understand · Apply

## Learning goals

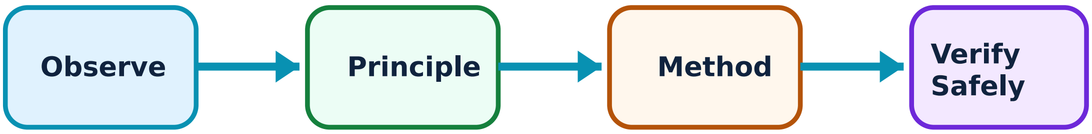
Identify core terms, explain the principle and complete the stated application or practical outcome.

## Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

## Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.




Study flow for Welding Shop: identify the system, state its principle, follow the correct method and verify the result safely.

### 1. Welding safety, PPE and setup−

Welding produces heat, intense light, fumes, spatter and electrical hazards. Safe practice needs helmet, gloves, clothing, screen, ventilation and inspection of leads/equipment.

#### Study and application method

Prepare clean joint, set up under supervision and maintain fire precautions. Explain why PPE protects eyes, skin and hands.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

### Practical method

1. Read the aim and identify apparatus, materials and workspace controls.
2. Prepare a labelled diagram, tabulation sheet or job plan before execution.
3. Carry out the prescribed operation methodically and record observations while working.
4. Verify the result against the aim, clean the work area and complete the record.

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**Common mistake / precaution:** Never look at arc without eye protection or weld near flammable material.

## 2. Arc, resistance and cabinet joints−

Official practice includes straight-line welding, spot/resistance welding and joining simple cabinets. Joint quality depends on fit-up, process selection and controlled heat input.

### Study and application method

Practise bead control on scrap then inspect for continuity, spatter, undercut or incomplete joining. Record process/material/inspection.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

### Practical method

1. Read the aim and identify apparatus, materials and workspace controls.
2. Prepare a labelled diagram, tabulation sheet or job plan before execution.
3. Carry out the prescribed operation methodically and record observations while working.
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**Common mistake / precaution:** Allow work to cool safely; hot metal may not appear hot.

**Module 4 revision cue:** Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 5

## Sheet Metal Work

Sheet-metal tools; cutting, bending, edging; fasteners; fabrication of casings for power supplies, inverters and UPS.

**18 practical hours** | **CO4** | **Understand · Apply**

### Learning goals

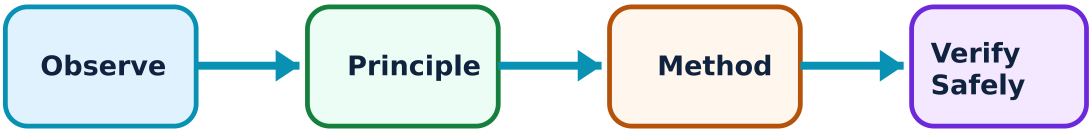
Identify core terms, explain the principle and complete the stated application or practical outcome.

### Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

## Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.




Study flow for Sheet Metal Work: identify the system, state its principle, follow the correct method and verify the result safely.

### 1. Sheet-metal operations and joints−

Sheet-metal work uses measurement, layout, cutting, bending, edging and joining. Snips, folding tools and supports are selected by shape and thickness.

## Study and application method

Mark development clearly, retain bend allowance as instructed and deburr after cutting. Practise joints before final casing.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

## Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

## Practical method

1. Read the aim and identify apparatus, materials and workspace controls.
2. Prepare a labelled diagram, tabulation sheet or job plan before execution.
3. Carry out the prescribed operation methodically and record observations while working.
4. Verify the result against the aim, clean the work area and complete the record.

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> **Common mistake / precaution:** Sharp edges can cause serious cuts; keep fingers clear of snips.

## 2. Electronic-equipment casing fabrication−

Official outputs include power-supply, inverter and UPS casings and fastening techniques. An enclosure needs correct dimensions, safe edges, assembly access and reliable fastening.

### Study and application method

Translate drawing to flat pattern, verify hole positions before drilling and inspect assembled alignment. Record tools, sequence and quality checks.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

### Practical method

1. Read the aim and identify apparatus, materials and workspace controls.
2. Prepare a labelled diagram, tabulation sheet or job plan before execution.
3. Carry out the prescribed operation methodically and record observations while working.
4. Verify the result against the aim, clean the work area and complete the record.

> **Malayalam support:** □□□□□ concept □□□□□□□□ □□□□□ □□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

**Common mistake / precaution:** A casing contributes to electrical safety; do not leave burrs or uncontrolled openings near live wiring.

**Module 5 revision cue:** Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 6

# Electrical Wiring and Basic Electronics

Electrical safety; one-way/staircase wiring; DOL starter; passive/active parts; rectifier/filter and Arduino familiarisation; soldering.

18 practical hours | CO5 | Understand · Apply

### Learning goals

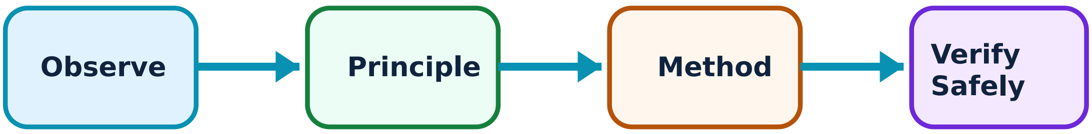
Identify core terms, explain the principle and complete the stated application or practical outcome.

### Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

### Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.




Study flow for Electrical Wiring and Basic Electronics: identify the system, state its principle, follow the correct method and verify the result safely.

### 1. Electrical safety, wiring and motor connection−

Official work includes one-lamp one-switch, staircase control and single-phase motor connection with DOL starter. Correct phase switching, insulation, earthing and isolation are fundamental.

#### Study and application method

Draw circuit/layout diagrams before wiring. Verify supply isolation before changes and request inspection before energising.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

#### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

#### Practical method

1. Read the aim and identify apparatus, materials and workspace controls.
2. Prepare a labelled diagram, tabulation sheet or job plan before execution.
3. Carry out the prescribed operation methodically and record observations while working.
4. Verify the result against the aim, clean the work area and complete the record.

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**Common mistake / precaution:** Do not work on live conductors or bypass fuse/protection device.

### 2. Components, rectifiers, Arduino and soldering−

Passive/active components, rectifier/filter circuits, Arduino UNO familiarisation and simple-circuit soldering are included. Soldering gives electrical/mechanical joint when heat, surface preparation and timing are controlled.

### Study and application method

Identify component markings/polarity. Heat joint rather than melting solder on iron tip alone; inspect for bridges/cold joints.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

### Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

### Practical method

1. Read the aim and identify apparatus, materials and workspace controls.
2. Prepare a labelled diagram, tabulation sheet or job plan before execution.
3. Carry out the prescribed operation methodically and record observations while working.
4. Verify the result against the aim, clean the work area and complete the record.

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**Common mistake / precaution:** Use fume control and return soldering iron to stand.

**Module 6 revision cue:** Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

## Rapid Recall Bank

### Recall 1

State symbols and SI units before substitution.

### Recall 2

> Draw the circuit, system boundary, vector or process flow before reasoning.

### Recall 3

> For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

### Recall 4

> For a graph: label axes, units, scale, operating region and conclusion.

### Recall 5

> For a comparison: use construction, working, result, advantage and limitation.

### Recall 6

> For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

## Diagram Library for Rapid Revision

### Module 1: Industrial Safety and Fire Practice

Use the concept flow to revise observation, principle, method and verification.

### Module 2: Carpentry and Wood Alternatives

Use the concept flow to revise observation, principle, method and verification.

### Module 3: Fitting Shop

Use the concept flow to revise observation, principle, method and verification.

### Module 4: Welding Shop

Use the concept flow to revise observation, principle, method and verification.

### Module 5: Sheet Metal Work

Use the concept flow to revise observation, principle, method and verification.

### Module 6: Electrical Wiring and Basic Electronics

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

## Official Self-Learning and Student Activities

### Structured activity

Maintain workshop records with labelled sketches, tool-purpose tables, procedure sheets and inspection notes.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

### Structured activity

Complete carpentry, fitting, welding, sheet-metal and electrical/electronic self-learning activities in official weekly sequence.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

### Structured activity

Undertake one open-ended experiment or prototype in student group as directed.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

**Activity discipline:** The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher's approved schedule and safety instructions govern execution.

EXAMINATION READINESS

## Assessment Methodology and Examination Pattern

| Official assessment structure | | |
|---|---|---|
| **Component** | **Official mark / conversion** | **Student evidence** |
| Attendance | 5 marks | Regular attendance and punctuality. |
| Continuous evaluation | Practical CIE 60 | Preparation, setup, observation, analysis, viva, record and discipline. |
| Practical tests | As stated in supplied PDF | Procedure/write-up, execution, result, viva and certified record. |
| Open-ended experiment | Where listed in supplied PDF | Originality, plan, execution/data, analysis/presentation and teamwork. |
| End-semester assessment | Practical ESE 40 | Safe completion, record and viva according to official scheme. |

### Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

### Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

# Module-wise Questions with Answers

## Module 1 — Industrial Safety and Fire Practice

**Explain Safe workshop behaviour and 5S and state one correct method, application or precaution.−**

Safe work begins before machine start: assess job, wear appropriate PPE, inspect tools, keep walkways clear and follow authorised procedures. 5S supports sorting, arranging, cleaning, standardising and sustaining disciplined workspace.

**Answer framework:** Pause before work to identify hazard, control and emergency response. Return tools to correct location and report damaged equipment.

**Important point:** Never operate equipment after removing guard or while distracted.

**Explain Fire classes and extinguishers and state one correct method, application or precaution.−**

Fire control depends on fuel type and extinguishing method. Official syllabus requires recognition of fire classes, matching extinguisher use and supervised hands-on practice.

**Answer framework:** Raise alarm, follow local instruction, maintain safe escape route and use extinguisher only when trained and fire is small/controllable.

**Important point:** Do not use water indiscriminately on electrical or flammable-liquid fires.

## Module 2 — Carpentry and Wood Alternatives

**Explain Tools, materials and preparation and state one correct method, application or precaution.−**

Carpentry accuracy depends on selecting measuring, marking, cutting, holding and striking tools for intended operation. MDF, WPC and plywood are official alternative materials and differ from solid timber in working behaviour.

**Answer framework:** Study board orientation, mark from datum edge and secure work before cutting. Record tool name, use and safety precaution.

**Important point:** Do not cut unsupported work or place hands in saw line.

**Explain Wood-panel box fabrication and state one correct method, application or precaution.−**

The official activity develops skill in cutting/drilling panels, making a simple box, and fixing hinges/handles. Accuracy comes from clear drawing, square assembly and final inspection.

**Answer framework:** Sequence drawing, measurement, marking, cutting, trial fit, drilling, fastening, inspection and finish. Check diagonals/corner squareness before final fastening.

**Important point:** Power tools require eye protection, correct clamping and instructor-approved settings.

## Module 3 — Fitting Shop

**Explain Fitting tools and measurement and state one correct method, application or precaution.−**

Fitting uses measured metal removal by marking, cutting, filing, drilling and finishing. Vernier calipers and micrometers are precision instruments, not rough layout tools.

**Answer framework:** Read zero, hold instrument square to dimension and avoid overtightening. Mark from datum and centre-punch drilling positions.

**Important point:** A file must have fitted handle; never use a file as lever or strike it with hammer.

**Explain Heat-sink, clamp and bracket fabrication and state one correct method, application or precaution.−**

Official applications require aluminium heat sinks or fabrication clamps/brackets, then drilling, screwing and fixing. Parts need functional dimensions and acceptable finish.

**Answer framework:** Use drawing for hole positions, edge distances and bends. Deburr every sharp edge and verify fit against intended panel before submission.

**Important point:** Freshly cut or ground metal edges can cause injury.

## Module 4 — Welding Shop

**Explain Welding safety, PPE and setup and state one correct method, application or precaution.−**

Welding produces heat, intense light, fumes, spatter and electrical hazards. Safe practice needs helmet, gloves, clothing, screen, ventilation and inspection of leads/equipment.

**Answer framework:** Prepare clean joint, set up under supervision and maintain fire precautions. Explain why PPE protects eyes, skin and hands.

**Important point:** Never look at arc without eye protection or weld near flammable material.

**Explain Arc, resistance and cabinet joints and state one correct method, application or precaution.−**

Official practice includes straight-line welding, spot/resistance welding and joining simple cabinets. Joint quality depends on fit-up, process selection and controlled heat input.

**Answer framework:** Practise bead control on scrap then inspect for continuity, spatter, undercut or incomplete joining. Record process/material/inspection.

**Important point:** Allow work to cool safely; hot metal may not appear hot.

## Module 5 — Sheet Metal Work

**Explain Sheet-metal operations and joints and state one correct method, application or precaution.−**

Sheet-metal work uses measurement, layout, cutting, bending, edging and joining. Snips, folding tools and supports are selected by shape and thickness.

**Answer framework:** Mark development clearly, retain bend allowance as instructed and deburr after cutting. Practise joints before final casing.

**Important point:** Sharp edges can cause serious cuts; keep fingers clear of snips.

**Explain Electronic-equipment casing fabrication and state one correct method, application or precaution.−**

Official outputs include power-supply, inverter and UPS casings and fastening techniques. An enclosure needs correct dimensions, safe edges, assembly access and reliable fastening.

**Answer framework:** Translate drawing to flat pattern, verify hole positions before drilling and inspect assembled alignment. Record tools, sequence and quality checks.

**Important point:** A casing contributes to electrical safety; do not leave burrs or uncontrolled openings near live wiring.

### Module 6 — Electrical Wiring and Basic Electronics

**Explain Electrical safety, wiring and motor connection and state one correct method, application or precaution.−**

Official work includes one-lamp one-switch, staircase control and single-phase motor connection with DOL starter. Correct phase switching, insulation, earthing and isolation are fundamental.

**Answer framework:** Draw circuit/layout diagrams before wiring. Verify supply isolation before changes and request inspection before energising.

**Important point:** Do not work on live conductors or bypass fuse/protection device.

**Explain Components, rectifiers, Arduino and soldering and state one correct method, application or precaution.−**

Passive/active components, rectifier/filter circuits, Arduino UNO familiarisation and simple-circuit soldering are included. Soldering gives electrical/mechanical joint when heat, surface preparation and timing are controlled.

**Answer framework:** Identify component markings/polarity. Heat joint rather than melting solder on iron tip alone; inspect for bridges/cold joints.

**Important point:** Use fume control and return soldering iron to stand.

PRACTICE ONLY

## Practice Model Paper

**Important:** This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

### Part A — short response

1. State one key definition from Module 1: Industrial Safety and Fire Practice.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: Carpentry and Wood Alternatives.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.

5. State one key definition from Module 3: Fitting Shop.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: Welding Shop.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.
9. State one key definition from Module 5: Sheet Metal Work.
10. Identify one diagram, observation, formula or safety condition relevant to Module 5.
11. State one key definition from Module 6: Electrical Wiring and Basic Electronics.
12. Identify one diagram, observation, formula or safety condition relevant to Module 6.

### Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.
5. Explain a selected procedure/principle from Module 5 with a labelled diagram, table or method.
6. Explain a selected procedure/principle from Module 6 with a labelled diagram, table or method.

### Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Industrial safety, hazardous situations, personal protection, fire classes and hands-on fire-extinguisher practice..
2. Develop a complete answer or practical plan for: Woodworking tools; MDF, WPC and plywood; marking, planing, cutting, chiselling and grooving; wooden-panel fabrication..
3. Develop a complete answer or practical plan for: Fitting tools and operations; heat sinks, clamps and brackets; drilling, screwing and precision measurement..
4. Develop a complete answer or practical plan for: Welding safety; equipment and PPE; straight-line/spot welding; joining simple cabinets..
5. Develop a complete answer or practical plan for: Sheet-metal tools; cutting, bending, edging; fasteners; fabrication of casings for power supplies, inverters and UPS..
6. Develop a complete answer or practical plan for: Electrical safety; one-way/staircase wiring; DOL starter; passive/active parts; rectifier/filter and Arduino familiarisation; soldering..

LAST-MILE REVISION

## Quick Revision

### Module 1: Industrial Safety and Fire Practice

Industrial safety, hazardous situations, personal protection, fire classes and hands-on fire-extinguisher practice.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

### Module 2: Carpentry and Wood Alternatives

Woodworking tools; MDF, WPC and plywood; marking, planing, cutting, chiselling and grooving; wooden-panel fabrication.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

### Module 3: Fitting Shop

Fitting tools and operations; heat sinks, clamps and brackets; drilling, screwing and precision measurement.

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

### Module 4: Welding Shop

Welding safety; equipment and PPE; straight-line/spot welding; joining simple cabinets.

- Match each topic to CO3.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

### Module 5: Sheet Metal Work

Sheet-metal tools; cutting, bending, edging; fasteners; fabrication of casings for power supplies, inverters and UPS.

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

### Module 6: Electrical Wiring and Basic Electronics

Electrical safety; one-way/staircase wiring; DOL starter; passive/active parts; rectifier/filter and Arduino familiarisation; soldering.

- Match each topic to CO5.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

**Common mistakes:** Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

**Exam checklist:** Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

## Glossary

| Important terms from the syllabus | |
|---|---|
| **Term / topic** | **One-line meaning** |
| Safe workshop behaviour and 5S | Safe work begins before machine start: assess job, wear appropriate PPE, inspect tools, keep walkways clear and follow authorised procedures. |
| Fire classes and extinguishers | Fire control depends on fuel type and extinguishing method. |
| Tools, materials and preparation | Carpentry accuracy depends on selecting measuring, marking, cutting, holding and striking tools for intended operation. |
| Wood-panel box fabrication | The official activity develops skill in cutting/drilling panels, making a simple box, and fixing hinges/handles. |
| Fitting tools and measurement | Fitting uses measured metal removal by marking, cutting, filing, drilling and finishing. |
| Heat-sink, clamp and bracket fabrication | Official applications require aluminium heat sinks or fabrication clamps/brackets, then drilling, screwing and fixing. |
| Welding safety, PPE and setup | Welding produces heat, intense light, fumes, spatter and electrical hazards. |
| Arc, resistance and cabinet joints | Official practice includes straight-line welding, spot/resistance welding and joining simple cabinets. |
| Sheet-metal operations and joints | Sheet-metal work uses measurement, layout, cutting, bending, edging and joining. |
| Electronic-equipment casing fabrication | Official outputs include power-supply, inverter and UPS casings and fastening techniques. |
| Electrical safety, wiring and motor connection | Official work includes one-lamp one-switch, staircase control and single-phase motor connection with DOL starter. |
| Components, rectifiers, Arduino and soldering | Passive/active components, rectifier/filter circuits, Arduino UNO familiarisation and simple-circuit soldering are included. |

LEARNING RESOURCES

# References

### Officially listed print resources


1. S. K. Hajara Chaudhary, Workshop Technology.
2. J. B. Gupta, Electrical Installation Estimation and Costing.
3. B. S. Raghuwanshi, Workshop Technology.

### Officially listed web resources

- https://bharatskills.gov.in/Home/StudyMaterial?var=wyt2HdOKOUuq3xZOoZ+r+w==&Default=No

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